



Data Communication Wireless Networks (IE2100)

2nd Year, 2nd Semester

Assignment

LAN & WLAN Network Design Proposal

Submitted to

Sri Lanka Institute of Information Technology

In partial fulfillment of the requirements for the
Bachelor of Science Special Honors Degree in Information Technology

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<u>Student Name</u>	<u>Reg Number</u>
<u>Minthaka M.M.</u>	<u>IT24102219</u>
<u>Samarasekara S.M.A.P.</u>	<u>IT24102353</u>
<u>Jayathunga H.P.K.S.</u>	<u>IT24101150</u>
<u>Bandara G.R.S.S.</u>	<u>IT24102229</u>

Project Plan

1. Project Overview

This project proposes the design, supply, installation, configuration, testing, documentation, and support of a modern wired and wireless network for Behpaya Information Technology's two-floor head office in Sri Lanka.

The solution is designed to provide a secure, reliable, scalable, and centrally managed network that supports approximately 50 staff members, up to 30 concurrent guest devices, and around 120 active endpoints over a 300 Mbps fiber connection. The project must cover wired LAN, wireless LAN, security, structured cabling, IP addressing, VLAN segmentation, centralized monitoring, configuration backup, and administrator access control.

2. Project Aim

The main aim of this project is to deliver a complete technical and financial proposal for a business-grade LAN and WLAN environment that can support current operational needs and future growth over the next 3 to 5 years.

The proposed solution must also satisfy the required deliverables such as low-level design, Wi-Fi design package, as-built diagrams, test and acceptance reports, configuration backup procedures, and administrator handover documentation

Project Methodology

Phase 1 – Requirement Analysis

In this phase, the business and technical requirements of Behpaya Information Technology will be studied. The office layout, number of users, expected devices, wireless coverage requirements, security requirements, and network management expectations will be identified.

Special attention will be given to the need for managed switches, PoE support, VLAN segmentation, Wi-Fi 6/6E access points, centralized monitoring, secure guest access, VPN support, and next-generation firewall capabilities.

Phase 2 – Site and Capacity Assessment

The office environment will be assessed using the given floor summary and capacity assumptions. The design will consider the ground floor areas such as reception, meeting rooms, open office, and café/lobby, as well as the first-floor areas including the open office, boardroom, printer area, and MDF/network closet.

Capacity planning will be based on 50 staff users, up to 30 guest users, and approximately 120 active endpoints. Coverage and performance targets such as at least 95% usable area coverage at -65 dBm or better will also be considered.

Phase 3 – High-Level Network Design

A high-level architecture will be prepared to show how the ISP connection, firewall, core switch, access switches, wireless access points, controller/cloud management platform, and optional internal services such as DHCP, DNS, NTP, and RADIUS will interact. At this stage, the network will be logically divided into required segments such as Staff, Admin, Guest, Network Devices, and optional Servers/IoT VLANs. The design will also show uplink structure, aggregation, and general security zones.

Phase 4 – Detailed Network Design

In this phase, the low-level design will be created. This includes the IP addressing plan, VLAN schema, DHCP scopes, SSID mapping, security policies, naming conventions, and link design. Detailed wired and wireless design decisions will be made, including access switch distribution, trunk links, AP placement, channel and power planning, roaming support, and link aggregation. Security measures such as 802.1X, DHCP snooping, dynamic ARP inspection, VPN access, and firewall policies will also be defined.

Phase 5 – Financial Proposal and Device Selection

After the technical design is finalized, an equipment list will be prepared based on the project requirements. The bill of materials will include firewall, switches, wireless access points, controller or cloud management platform, rack, UPS, patch panels, cables, and other supporting items. A financial proposal will then be developed to justify the selected solution in terms of cost, suitability, scalability, and manageability. This phase ensures that the recommended network is not only technically strong but also economically reasonable for the client.

Phase 6 – Implementation Planning

A practical implementation approach will be prepared to show how the network would be deployed if approved. This includes structured cabling work, rack setup, device installation, VLAN configuration, firewall policy setup, wireless controller configuration, AP adoption, SSID creation, monitoring setup, and configuration backup preparation. This phase will also define the order of deployment to minimize disruption and ensure successful commissioning.

Phase 7 – Testing and Validation

The proposed network will then be validated against the required acceptance criteria. Testing will include VLAN connectivity checks, DHCP assignment, internet access validation, guest isolation testing, wireless coverage validation, throughput verification, firewall rule checks, VPN verification, and monitoring alert tests. Coverage KPIs, failover behavior if applicable, and security policy verification will be included as part of the final test and acceptance report.

Phase 8 – Documentation and Handover

The final stage will produce all required documents including low-level design documents, predictive heatmaps, AP placement diagrams, as-built diagrams, device inventory, configuration backup procedures, and quick-start administrator guidance. This stage ensures that the client can manage, maintain, and expand the network after project completion.

Resource Allocation

The report is prepared as a technical and financial network design document for Behpaya Information Technology, covering wired LAN, wireless LAN, security, network services, monitoring, documentation, and presentation preparation.

To ensure that all required sections are completed efficiently and to a good standard, the project work is divided among the four group members based on major technical areas in the assignment. This allocation supports the required deliverables such as the low-level design, Wi-Fi planning package, device inventory, security policy set, test and acceptance report, and final presentation.

Member ID	Assigned Role	Main Responsibilities
IT24102219	Group Coordinator and Wired Configuration Support	Diagram creation, wired configuration, final testing, and coordination of group work
IT24101150	Wireless Design and Architecture Lead	Wireless configuration, architecture designing, final testing, and simulation creation
IT24102229	Wireless and Simulation Support Lead	Wireless configuration, simulation creation, connecting devices, and final testing
IT24102353	Wired Planning, Heatmap, and Financial Report Lead	Wired configuration, predictive heatmap preparation, financial report creation, final testing, and floor map design

(Some tasks will be completed jointly to maintain consistency across the report.)

Project Timeline

Phase	Activity	Duration
Phase 1	Requirement analysis and site assessment	2 Days
Phase 2	High-level and low-level design preparation	2 Days
Phase 3	Device selection, budgeting, and approval	1 Days
Phase 4	Structured cabling, rack setup, and labeling	2 Days
Phase 5	Installation of firewall, switches, controller, and Aps	2 Days
Phase 6	VLAN, DHCP, SSID, firewall, VPN, and monitoring configuration	2 Days
Phase 7	Testing, validation, optimization, and security verification	2 Days
Phase 8	Documentation, backup handover, and admin guidance	1 Days

Risk & Communication

Effective risk management and clear communication are important to the success of this network design project. Since the assignment requires a complete technical and financial proposal for a modern wired and wireless network, the team must identify possible risks that could affect design quality, implementation feasibility, security, cost estimation, documentation, and presentation readiness. A proper communication plan is also necessary to ensure that all group members complete their assigned tasks on time and that the final report remains consistent and well organized.

Risk Management Approach

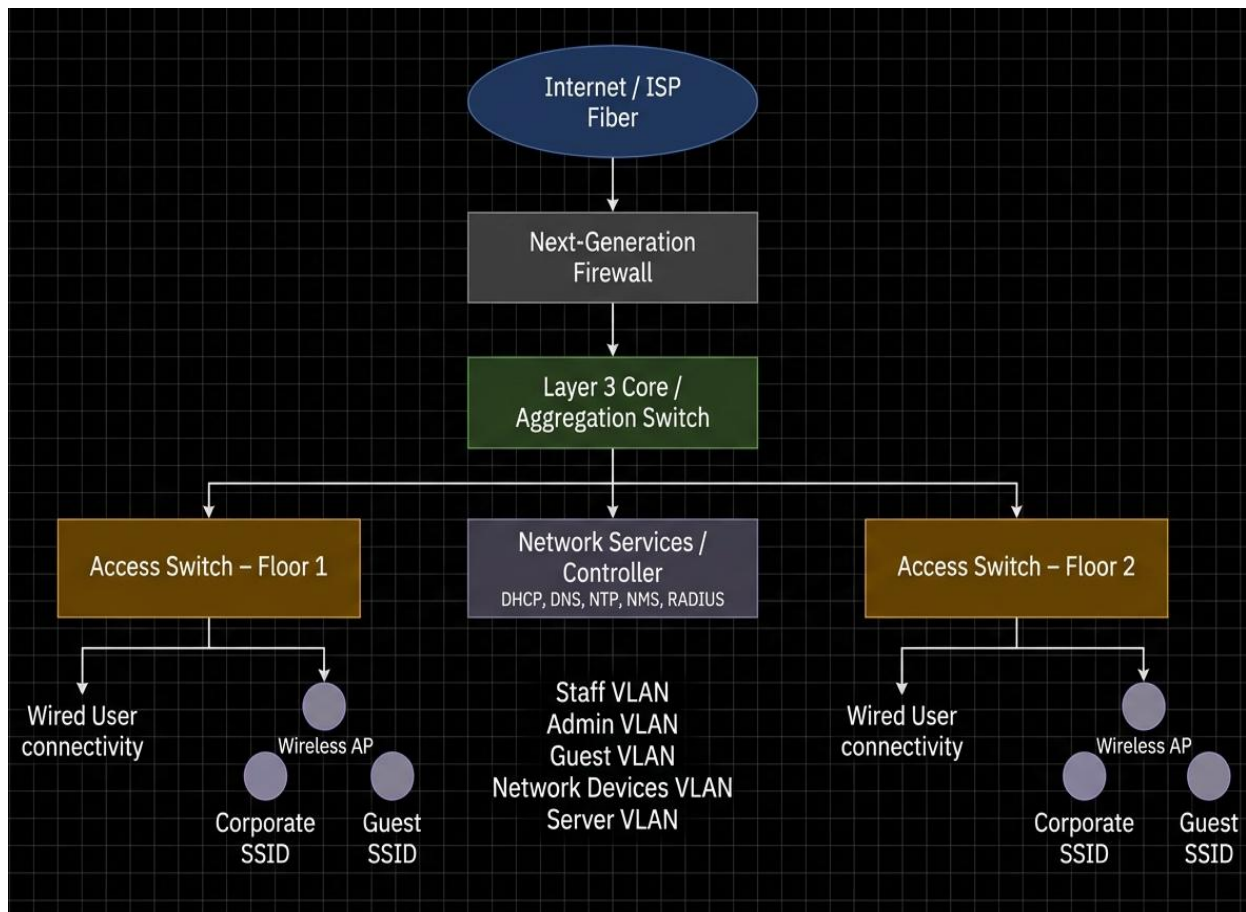
The project will use a simple risk management approach consisting of four steps,

1. Identify possible technical, project, and documentation risks
2. Assess the impact of each risk on the overall assignment
3. Define mitigation actions to reduce the likelihood or effect of the risk
4. Monitor progress through regular group communication and section review

Requirements

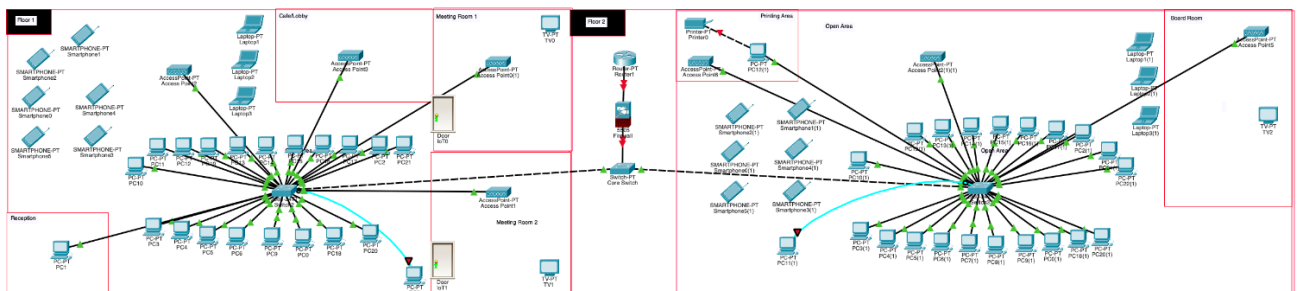
Requirement Area	Requirement from Brief
Office Size and Users	Two-floor office, 50 staff, 30 guest devices, 120 active endpoints
Wired LAN	Managed L2/L3 switches, PoE+, 1/10 Gbps uplinks, link aggregation
Segmentation	VLANs for Staff, Admin, Guest, Network Devices, optional Servers/IoT
Wireless LAN	Wi-Fi 6/6E APs, centralized control, RF planning, roaming optimization
Security	NGFW, IPS/IDS, URL filtering, VPN, NAC/policy-based segmentation
Network Services	DHCP, DNS, NTP, monitoring, backups, RBAC
Deliverables	LLD, heatmaps, diagrams, inventory, test report, handover

High-Level Architecture

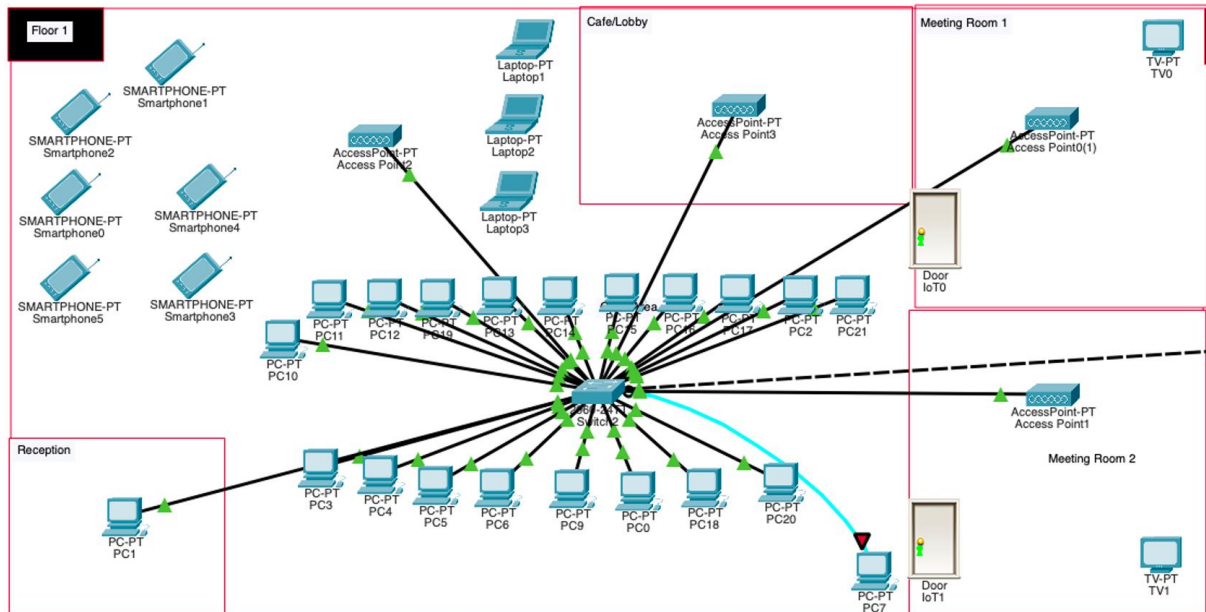


Proposed Wired and Wireless Network Layout for the Two-Floor Office

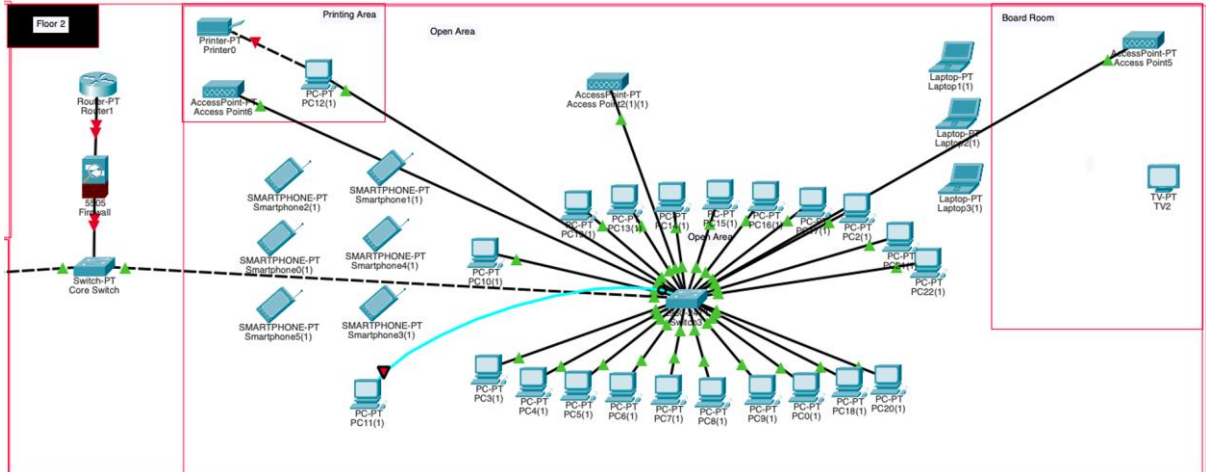
This diagram illustrates the overall network design for Behpaya Information Technology, showing the placement of core devices, access switches, wireless access points, and end-user devices across both floors. It highlights the integration of wired and wireless connectivity, room-based device distribution, and centralized network structure for secure and efficient office communication.



Floor 1



Floor 2



WLAN Coverage & RF Planning

The WLAN configuration shown in the floor layout uses seven access points distributed across the two floors of Behpaya Information Technology's office in order to provide room-based and user-density-based wireless coverage. Floor 1 contains four access points placed in the open office area, café/lobby, Meeting Room 1, and Meeting Room 2, while Floor 2 contains three access points placed in the printing area, main open area, and board room. This placement strategy ensures that both high-density working areas and enclosed rooms receive direct wireless coverage instead of depending only on signal spillover from nearby zones.

On Floor 1, Access Point2 is positioned to serve the main open office area where several wireless users are located. Access Point3 is placed in the café/lobby to provide dedicated coverage for that shared area. Access Point1 is located in Meeting Room 1, and Access Point1 is positioned in Meeting Room 2, giving both meeting rooms direct Wi-Fi coverage for video conferencing, collaboration, and mobile device connectivity. On Floor 2, Access Point2 covers the main open area, Access Point6 serves the printing area, and Access Point5 provides dedicated coverage to the board room. This arrangement indicates that the design has been planned according to both room separation and expected user concentration.

SSID & VLAN Mapping

The proposed wireless network will use a structured SSID-to-VLAN mapping model to separate employee, administrative, guest, and infrastructure traffic. This design improves network security, simplifies management, and ensures that different categories of users and devices are placed into the correct logical network segments. Based on the final addressing plan, five VLANs are defined: Staff, Admin, Guest, Network Devices, and Servers.

- **Behpaya-Corp → VLAN 10**
- **Behpaya-Admin → VLAN 20**
- **Behpaya-Guest → VLAN 30**

SSID to VLAN Mapping

SSID Name	VLAN ID	VLAN Name	Subnet	Gateway	DHCP Range
Behpaya-Corp	10	Staff	192.168.10.0/24	192.168.10.1	192.168.10.50 – 192.168.10.220
Behpaya-Admin	20	Admin	192.168.20.0/24	192.168.20.1	192.168.20.50 – 192.168.20.120
Behpaya-Guest	30	Guest	192.168.30.0/24	192.168.30.1	192.168.30.50 – 192.168.30.230

Infrastructure VLANs

VLAN ID	VLAN Name	Purpose	Subnet	Gateway	Addressing
40	Network Devices	Switches, access points, controller, firewall management	192.168.40.0/24	192.168.40.1	No DHCP, static only
50	Servers	RADIUS server, internal DNS, and optional internal services	192.168.50.0/24	192.168.50.1	No DHCP, static only

Static IP Allocation

Device / Service	IP Address
Core switch / routed side	192.168.254.1/30
Firewall internal uplink	192.168.254.2/30
Controller / NMS	192.168.40.10
Access switch 1	192.168.40.11
Access switch 2	192.168.40.12
Core switch management	192.168.40.13
AP 1	192.168.40.21
AP 2	192.168.40.22
AP 3	192.168.40.23
AP 4	192.168.40.24
AP 5	192.168.40.25
AP 6	192.168.40.26
AP 7	192.168.40.27
RADIUS server	192.168.50.10
Internal DNS / optional server	192.168.50.11

DNS Assignment

Internal DNS

- Primary internal DNS: 192.168.50.11

Public Backup DNS

- 8.8.8.8
- 1.1.1.1

VLAN Name	Primary DNS	Secondary DNS
Staff	192.168.50.11	8.8.8.8
Admin	192.168.50.11	8.8.8.8
Guest	8.8.8.8	1.1.1.1
Network Devices	192.168.50.11	Static only
Servers	192.168.50.11	Static only

WPA2/WPA3 & Authentication

The proposed wireless security design uses separate authentication and encryption methods for each SSID according to its purpose and security level.

Authentication Model

SSID Name	VLAN ID	VLAN Name	Security Mode	Authentication Method	Purpose
Behpaya-Corp	10	Staff	WPA2/WPA3-Enterprise	802.1X with RADIUS	Secure wireless access for employee devices
Behpaya-Corp	20	Admin	WPA2/WPA3-Enterprise	802.1X with RADIUS	Secure wireless access for admin and management users
Behpaya-Guest	30	Guest	WPA2/WPA3-Personal or Open with Captive Portal	Guest portal / temporary credentials	Internet-only access for visitors

VLAN & Switching Design

The proposed wired network is designed using a **segmented VLAN-based switching model** so that user devices, administrators, guests, infrastructure devices, and internal services can be separated into different logical networks.

VLAN Structure

VLAN ID	VLAN Name	Purpose	Subnet	Gateway	DHCP Range
10	Staff	Employee PCs, laptops, and normal office user devices	192.168.10.0/24	192.168.10.1	192.168.10.50 – 192.168.10.220
20	Admin	IT and administrative devices	192.168.20.0/24	192.168.20.1	192.168.20.50 – 192.168.20.120
30	Guest	Visitor internet-only access	192.168.30.0/24	192.168.30.1	192.168.30.50 – 192.168.30.230
40	Network Devices	Switches, APs, controller, and firewall management	192.168.40.0/24	192.168.40.1	Static only
50	Servers	RADIUS, DNS, and optional internal services	192.168.50.0/24	192.168.50.1	Static only

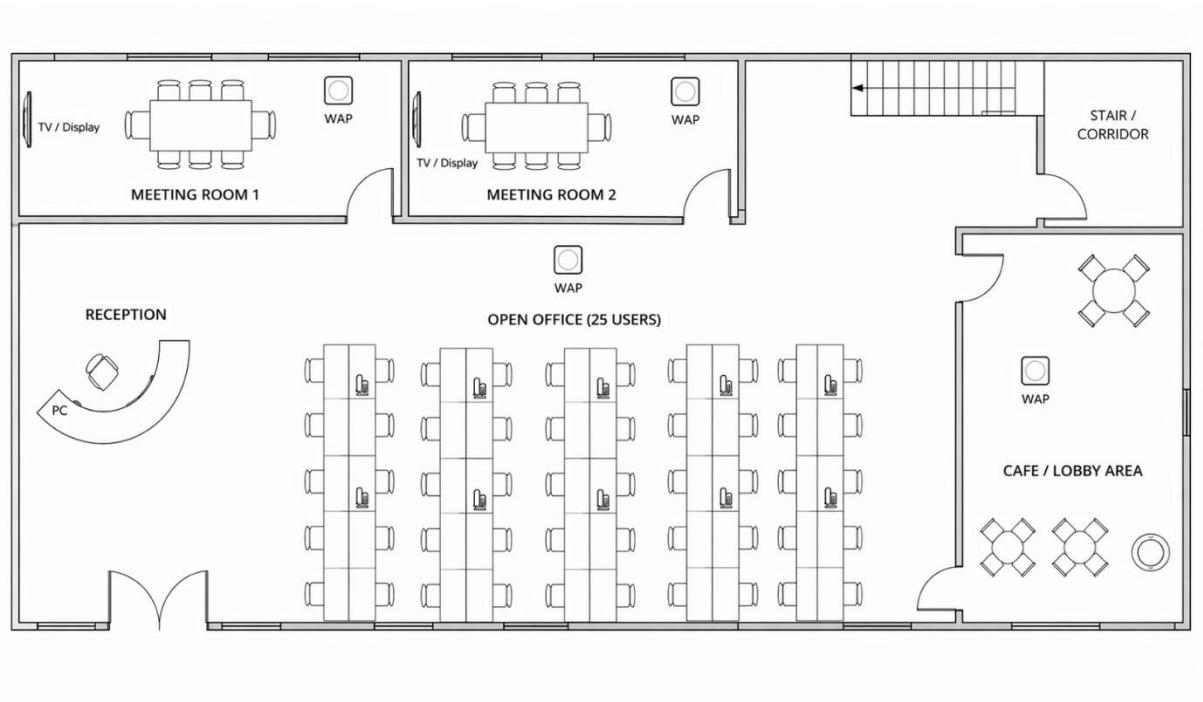
Link Layer Concepts

At the link layer, communication takes place using **Ethernet frames** between devices connected to the same local network. Switches make forwarding decisions based on **MAC addresses**, which are used to identify devices at Layer 2 of the OSI model. In the proposed design, access switches connect end devices such as PCs, printers, and wireless access points, while the core switch aggregates traffic and supports VLAN-based segmentation. This allows traffic to be delivered efficiently within each broadcast domain while maintaining logical separation between user groups.

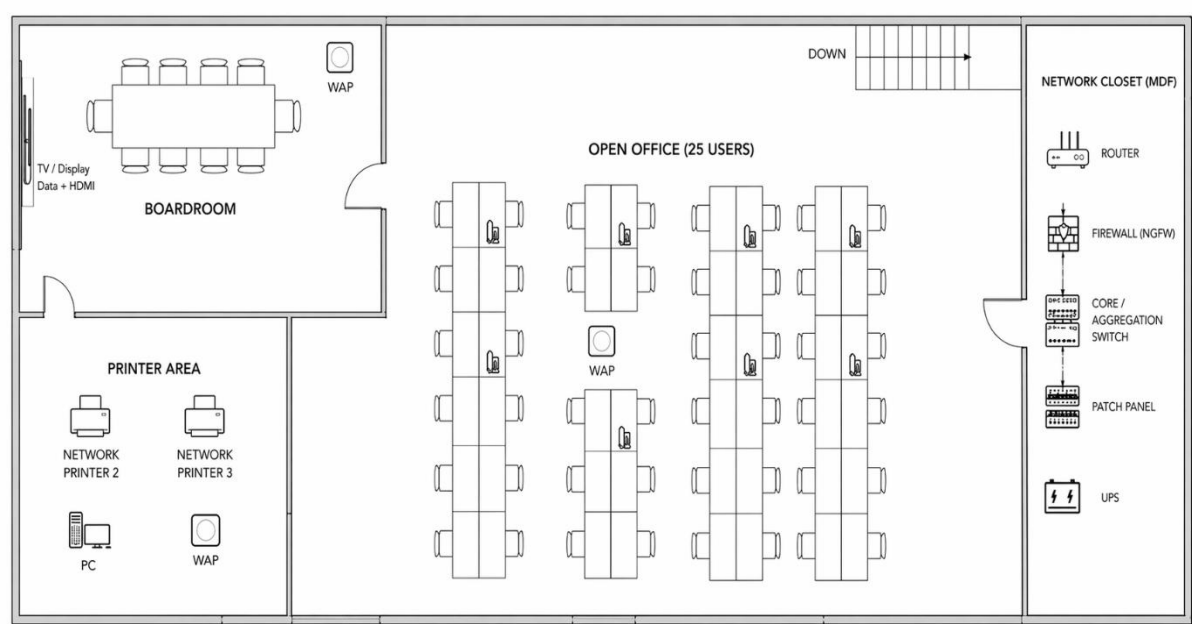
Real-World Mapping

The proposed network design for Behpaya Information Technology is developed to match the actual operational needs of a two-floor office environment rather than remaining only as a theoretical network model. The office includes a reception area, two meeting rooms, an open office and café/lobby on the ground floor, and an open office, boardroom, printer area, and MDF/network closet on the first floor. The design must also support approximately 50 staff members, up to 30 concurrent guest devices, and around 120 active endpoints while maintaining secure, reliable, and manageable connectivity

Ground Floor (first floor) Floor plan.

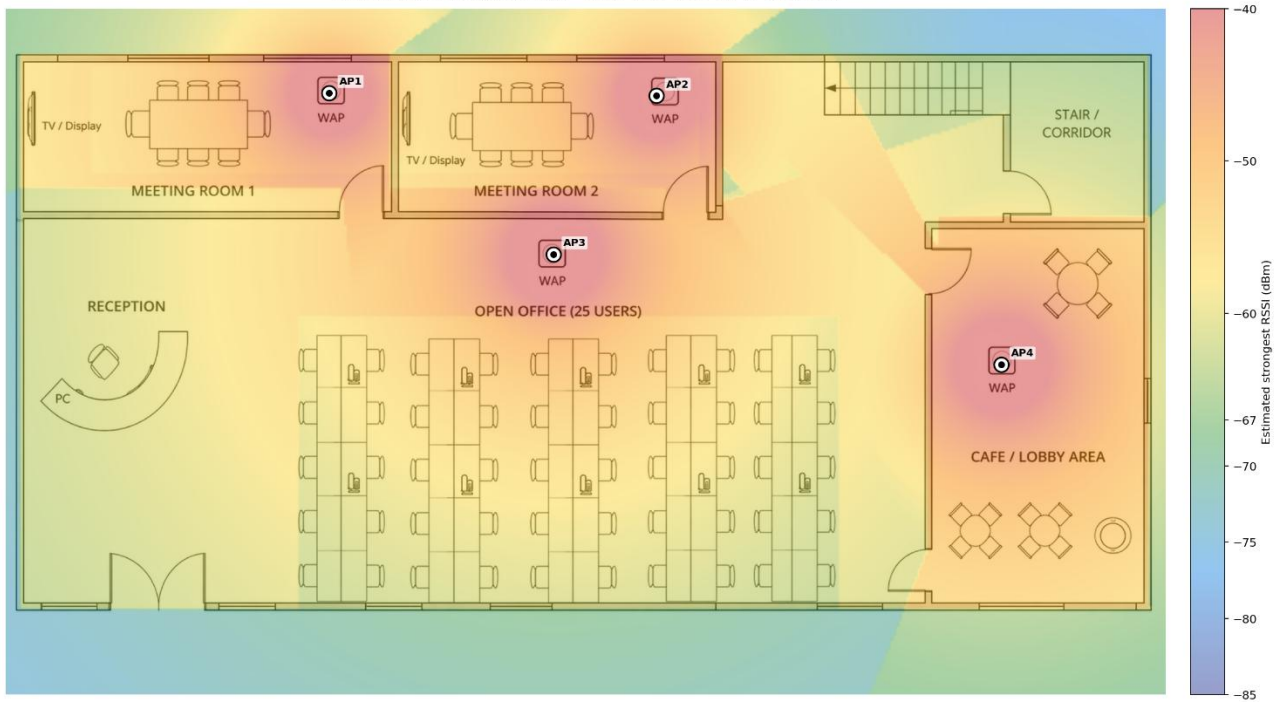


First Floor (second floor) Floor plan.



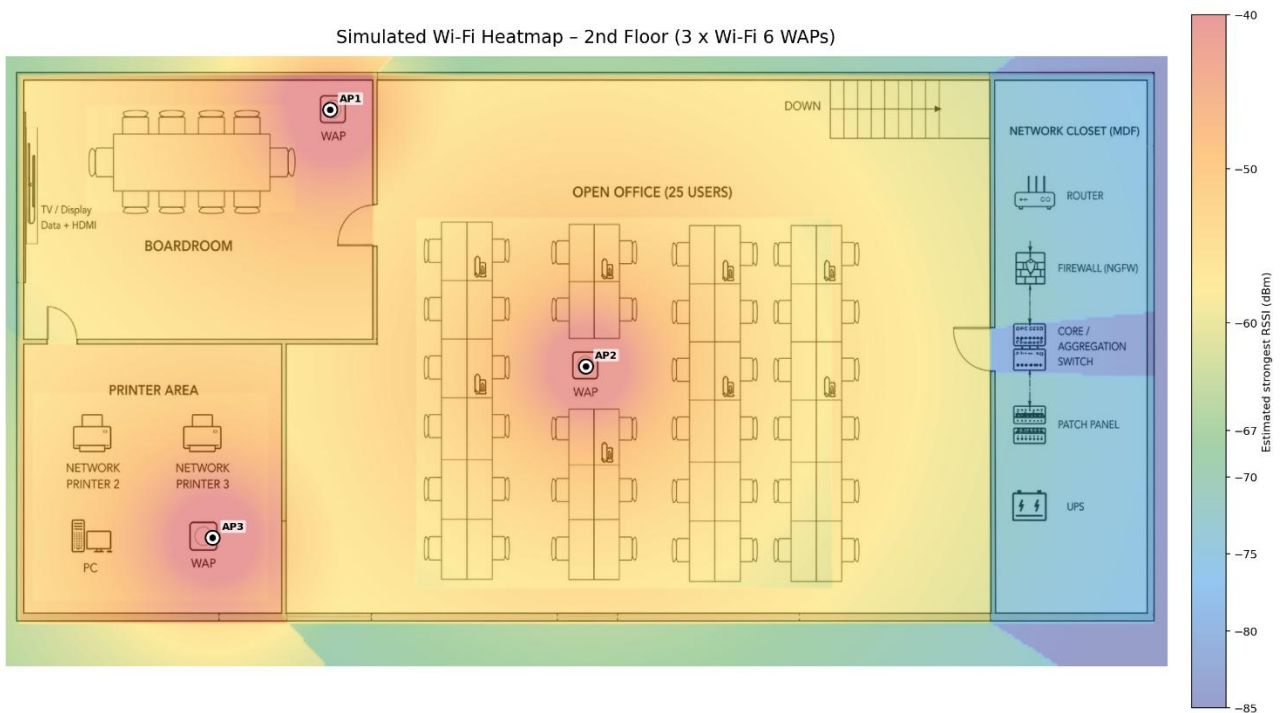
Predicted HeatMap (Ground Floor)

Simulated Wi-Fi Heatmap - 1st Floor (4 x Wi-Fi 6 WAPs)



Predicted HeatMap (First Floor)

Simulated Wi-Fi Heatmap - 2nd Floor (3 x Wi-Fi 6 WAPs)



Threat Identification

The proposed network faces several important threats, including unauthorized access, guest-to-internal network exposure, malware from the internet, rogue DHCP servers, ARP spoofing, weak VPN security, unprotected management access, wireless interception, and service outages caused by device failure or misconfiguration. These threats are relevant because the office requires secure wired and wireless access, guest connectivity, VPN support, centralized management, and reliable internet access. By identifying these risks early, the network can be designed with suitable protections such as VLAN segmentation, enterprise wireless authentication, firewall filtering, DHCP snooping, Dynamic ARP Inspection, role-based management access, and monitoring/backup procedures.

Security Implementation

The proposed security implementation uses a layered approach to protect the network. A next-generation firewall at the edge will provide IPS/IDS, URL filtering, malware protection, application control, VPN access, and traffic policy enforcement. Internally, the network is divided into separate VLANs for Staff, Admin, Guest, Network Devices, and Servers so that traffic is isolated according to role and function. The **Behpaya-Corp** and **Behpaya-Admin** SSIDs use **WPA2/WPA3-Enterprise with 802.1X and RADIUS**, while the **Behpaya-Guest** SSID is isolated and limited to internet-only access. Additional protections such as DHCP snooping, Dynamic ARP Inspection, role-based administrator access, monitoring, and configuration backups strengthen the overall security of the proposed office network.

NMS Features & Justification

The proposed network should include a centralized Network Management System to monitor switches, access points, uplinks, and key services from a single platform. Important features include device status monitoring, threshold-based alerts for WAN failure, AP offline status, and switch port errors, wireless monitoring, configuration backup, role-based administrator access, and reporting. These features are justified because the network includes multiple managed devices, VLANs, SSIDs, and security controls that would be difficult to manage manually. A centralized NMS improves visibility, reduces troubleshooting time, supports backup and recovery, and makes the overall network easier to operate and maintain.

Technical Detail

The design uses a hierarchical office network with an ISP connection, next-generation firewall, Layer 3 core switch, access switches, seven wireless access points, and centralized controller/NMS support. The network is divided into five VLANs: Staff, Admin, Guest, Network Devices, and Servers. Inter-VLAN routing is handled by the core switch, while the firewall applies internet and access-control policies. Static IP addresses are assigned to infrastructure devices, and DHCP is used for user VLANs. The wireless design uses three SSIDs mapped to the user VLANs, while switching uses access ports for end devices and trunk ports for uplinks and APs. Additional detail is provided through security controls, monitoring, naming conventions, and validation tests, giving the design practical implementation depth.

Financial Proposal / Cost Estimation

These values are approximate and may vary according to vendor pricing, product availability, and implementation conditions.

Item No.	Item Description	Quantity	Estimated Unit Cost (LKR)	Estimated Total (LKR)
1	Next-Generation Firewall with security services	1	420,000	420,000
2	Layer 3 Core Switch	1	180,000	180,000
3	48-Port PoE+ Managed Access Switch	2	275,000	550,000
4	Wi-Fi 6 / 6E Access Point	7	78,000	546,000
5	Controller / NMS Appliance or License	1	85,000	85,000
6	Rack Cabinet / Wall Mount Rack	1	95,000	95,000
7	UPS for network equipment	2	190,000	380,000
8	Patch Panels	4	12,000	48,000
9	Cat6 Cable and structured cabling materials	1 lot	160,000	160,000
10	Keystone jacks, faceplates, patch cords, labeling accessories	1 lot	90,000	90,000

Item No.	Item Description	Quantity	Estimated Unit Cost (LKR)	Estimated Total (LKR)
11	Installation and physical setup	1 lot	120,000	120,000
12	Configuration and system integration	1 lot	110,000	110,000
13	Testing, validation, and documentation support	1 lot	70,000	70,000
14	Training / handover support	1 lot	60,000	60,000

Estimated Subtotal

Cost Category	Amount (LKR)
Hardware and materials subtotal	2,554,000
Services subtotal	360,000
Project subtotal	2,914,000

Contingency and Final Total

Cost Category	Amount (LKR)
Project subtotal	2,914,000
Contingency reserve (10%)	291,400
Estimated grand total	3,205,400

(A 10% contingency amount is added to the budget to cover unforeseen costs)

Device Details

Item	Brand	Model
Firewall	Fortinet	FG-70F-BDL-809-12
Core Switch	Ubiquiti	USW-Pro-24
Access Switch	Ubiquiti	USW-48-PoE
Access Point	Ubiquiti	U6-Pro
Controller / NMS	Ubiquiti	UCK-G2-PLUS
Rack Cabinet	Generic / Local	9U Wall Mount Rack Cabinet
UPS	Alfa	Neo Plus-3000
Patch Panel	Red Link	RL-CAT6-24-P-PL
Cat6 Cable	3M	Cat-6 Full Box
Keystone Jack	AMP	CONAMP0003
Faceplate	AMP	RJ45 Face Plate Single AMP
Patch Cord	Vcom	NP611

Conclusion

The proposed network design provides a secure, reliable, and scalable wired and wireless solution for Behpaya Information Technology's two-floor office. The design meets the required technical needs through proper VLAN segmentation, structured IP addressing, centralized management, wireless coverage planning, and layered security controls. Overall, this solution is suitable for supporting daily business operations, guest access, and future organizational growth in an efficient and manageable way.